جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
نزوى
Nizwa

Information Technology Department

Final Examination
Fall, AY 2022/2023
CSCM1101 – Fundamentals of Computing and Information Systems

Date: Sun, Jan 15, 2023 **Time:** 9:00 AM – 11:00 AM

Max Marks	40	Time Allocation	2 Hours

Student ID		Student Name	
Section		Level	Diploma Year I

Instructions to the Students

1. Answer the questions in the space provided by using **blue or black pen** only. You will obtain a zero (0) if you use a pencil for all questions.
2. Use of mobile phones and any electronic gadgets inside the examination hall is not allowed.
3. ***Copying, Cheating, and any kind of malpractices on the examination are strictly prohibited.*** Action will be taken as per the "Article 80, Bylaws of Colleges of Technology".
4. Students should bring their own pencils, pens, erasers, rulers, calculators, and any other tools required for the exam.
5. The exam has a total of **24 non-optional questions** in **11 pages.**
6. Only the **Standard Calculator** is allowed during the exam.

Section	Allotted Marks	Obtained Marks
Section A	**10**	
Section B	**10**	
Section C	**14**	
Section D	**6**	
Total Marks	**40**	

Marked By	Checked/Re-marked by

Section A. Multiple Choice Questions. [10 x 1 = 10 marks]

Direction: Read the statement in each question and choose the best answer. Write the letter of your answer in the spaces provided below:

_____1. It refers to a data that have been organized so that they have a meaning and value to the recipient

a. Knowledge
b. Information
c. Information System
d. Input

_____2. It refers to a set of programs that enable users to perform specific information-processing activities.

a. Information System
b. Software
c. Systems Software
d. Application software

_____3. What are the primary production factors provided by the environment?

a. Input and Output
b. Hardware and Software resources
c. Capital and Labor
d. Data and Information

_____4. Dunlop Tires are selling their products and services online to Toyota using Electronic Data Interchange (EDI) via the Internet or extranet. Toyota uses the products they buy from Dunlop Tires as components to build the cars that they will sell to their customer.

Which category of e-commerce exists between Dunlop Tires and Toyota?

a. B2B
b. B2C
c. C2C
d. C2B

_____5. The output of a NAND gate ~(AB) where A=0 and B=1 is:

a. 0
b. 1
c. Error
d. No output

_____6. It refers to a number system where each symbol represents the same value regardless of its position.

a. Base
b. Digit
c. Positional
d. Non-positional

_____7. The codes for the following languages are interpreted except for:

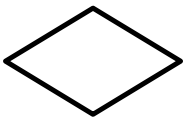
a. C++
b. Python
c. Ruby
d. Java

_____8. Suad wants to add a part in her flowchart that states" "Print Resident ID". Which flowcharting symbol will Suad use?

a.

b.

c.

d. 

_____9. Who are the IT professionals that work behind the scenes to make sure that all information systems we take for granted run smoothly?

a. Administrators
b. Systems Analysts
c. Hardware and Sofwre Engineers
d. Developers and Designers

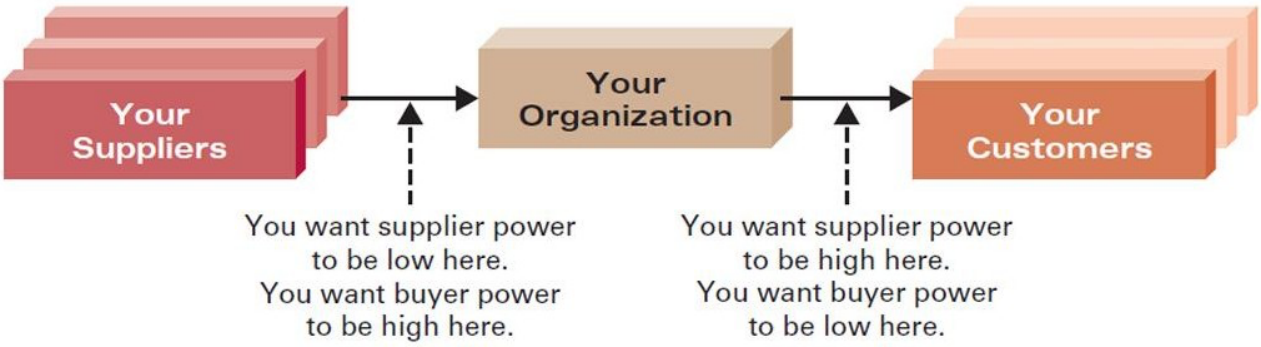
_____10. Mariyah wants to take a certification that covers installing and managing medium-sized networks. What certication should she take?

a. Cisco Certified Network Professional (CCNP)
b. Cisco Certified Network Associate (CCNA)
c. Microsoft Certified Systems Engineer (MSCE)
d. Oracle Certified Profesional (OCP)

Total for Section A
/10

Section B. Short Answer Questions. (5 x 2 = 10 marks)
Direction: Read each question and write your answers as required on the spaces provided.

11. Explain the Supplier Power diagram below: [2 marks]



Supplier Power refers to the ability of suppliers to influence the price of terms of supply.A high supplier power means the supplier has more control over the prices , quality and delivery.

12. List down the categories of e-commerce [2 marks]

Business to Business (B2B)

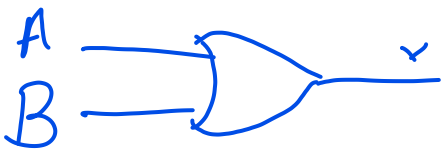
Business to consumer (B2C)

Consumer to consumer (C2C)

Consumer to Business(C2B)

13. What is an OR gate? Support your answer with the symbol and truth table. [2 marks]

if the two input values are both 0 , the output value is 0 otherwise the output is 1.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

14. Differentiate profit organization with non-profit organization. [2 marks]

15. When do you use a selection control structure? [2 marks]

Allows for a choice between two options or more that is allows you to make decision.

Total for Section B
/10

Section C. Application. [14 marks]

16. Convert the following numbers to the required number system [2 marks]

a. $EB8A_{16}$ = _____ $_{10}$

3 2 1 0

14 12 10

$14\times16^3 + 11\times16^2 + 8\times16^1 + 10\times16^0$

$57344 + 2816 + 128 + 10$

$= 60298$

b. 7213_8 = _____ $_{16}$

7 2 1 3

421 421 421 421

111 010 001 011

1110 1000 1011

8421 8421 8421

14 8 11

E 8 B

17. Perform binary addition [1 mark]

111

11010_2

$+\ 11110_2$

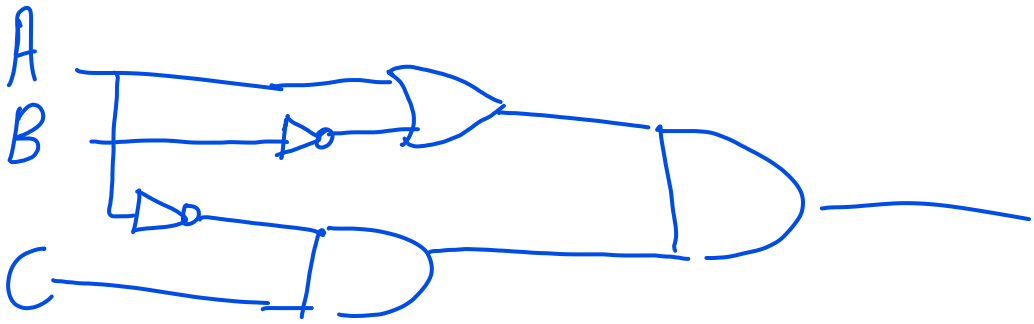
$\boxed{111000}$

18. Do the following for the given Boolean expression below:

a. **Draw the equivalent logic circuit [1.5 marks]**

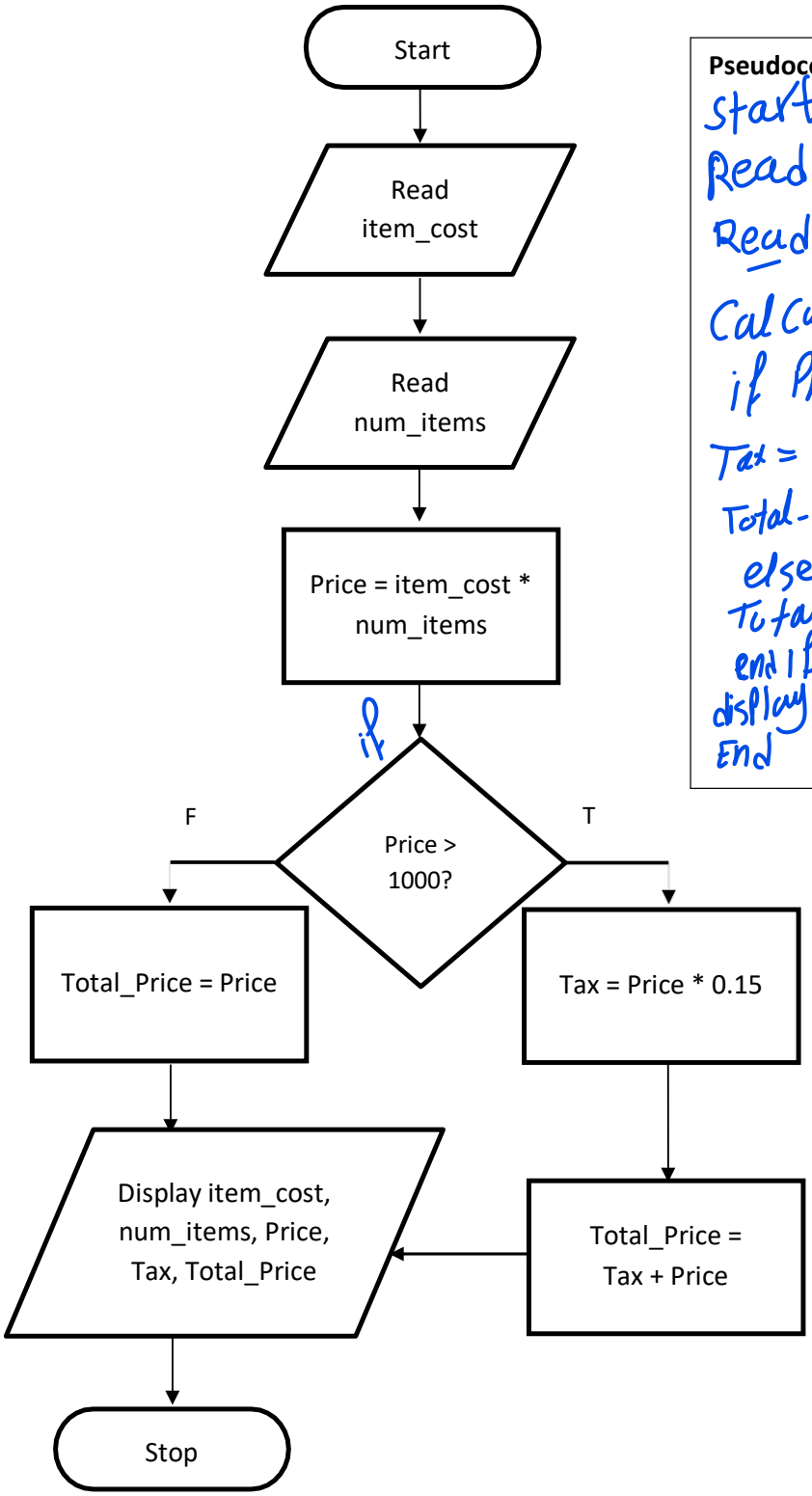
b. **What will be the output of the given logic circuit if A=1, B=1 and C=0? [1.5 marks]**

$$T = (A + B') \wedge (A'C)$$



A	B	C	A'	B'	(A + B')	(A' C)	T
1	1	0	0	0	1	0	0

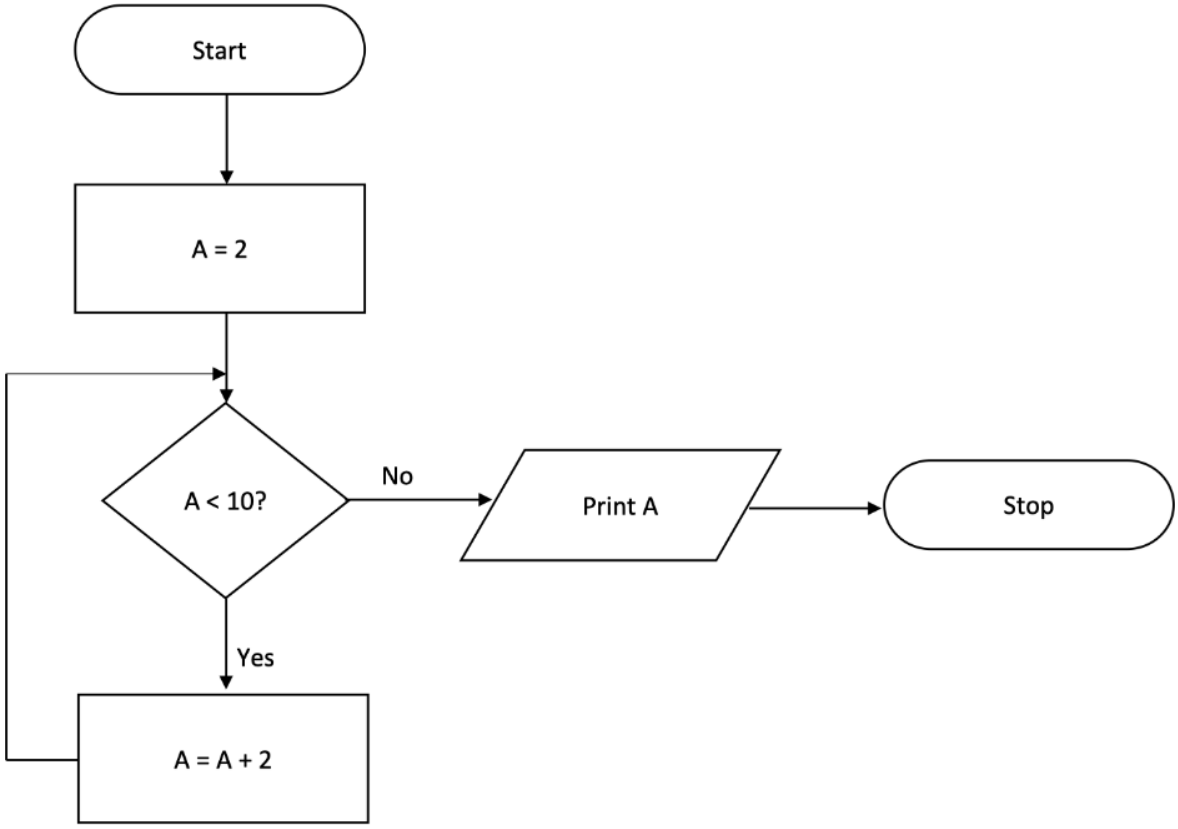
19. Convert the following flowchart to its equivalent pseudocode. You may write the answer in the space on the right side of the flowchart. [3 marks]



Pseudocode:

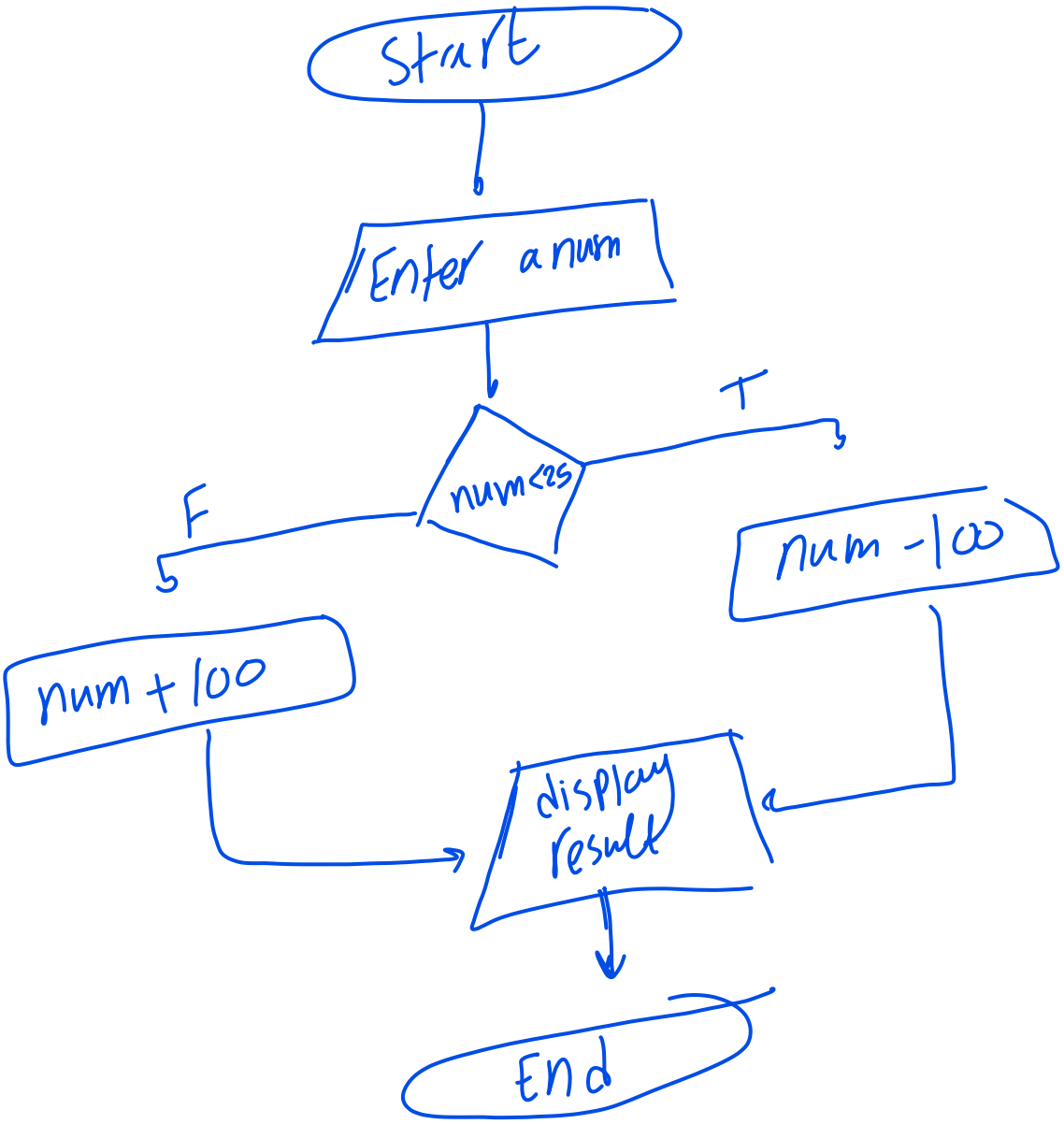
Start
Read item_cost
Read num_items
Calculate Price = item_Cost * num_item
if Price > 1000
Tax = Price * 0.15
Total_Price = Tax + Price
else
Total_Price = Price
end if
display item_Cost, num_item, Price, Tax, total_Price
End

20. Do the simulation of the following flowchart and find the output [2 marks]



21. Draw the flowchart for the following scenario: [3 marks]

The user will enter a number. If the number entered is less than 25, the number will be subtracted from 100, otherwise, the number will be added to 100. After the calculation, the Difference or Sum will be displayed.



Total for Section C
/14

Section D. Case Scenario Questions. (3 x 2 = 6 marks)
Direction: Read the case scenario below and answer following questions.

Glass Onion Company was established in 2006 when its owner Mr. Mohsin set up a small company that sells textiles in Nizwa Souq. The company became popular and towards the end of 2014, Mr. Mohsin expanded his business in alliance with Cotton & Silk Ltd. His company was able to produce low-cost textiles by getting raw materials directly from manufacturers.

In 2019, Glass Onion Company is considered as a very big organization with 600 employees all over Oman with 3 branches distributing textiles all over the country. The branch managers submit the sales report through the Glass Onion Management information System. The sales personnel use a tablet, which connects to a POS server through hubs, routers and fiber optics. It is the first textile company to use a POS for faster and efficient transaction.

The company launched its website to allow online ordering and direct selling to its customers inside and outside Oman. The customers need to register to the website before starting a transaction by entering the full name, email, and mobile number. The details will be verified and a One-Time PIN (OTP) will be sent to the mobile number to activate the account.

22. Identify the Input, Process and Output to register in the Glass Onion website [2 marks]

INPUT	PROCESS	OUTPUT
full name, email mobile number.	details verification OTP send	activite account

23. Identify one competitive strategy mentioned in the scenario and explain how it will help Glass Onion Company to gain a competitive advantage. [2 marks]

-getting row materials directly from manufacturers.

24. What is the e-commerce category described in the case scenario? Explain your answer. [2 marks]

B2C

Total for Section D
/6

****** End of Exam ******